

Transformers and Scaled Dot-Product Attention

A Technical Deep-Dive into Modern Neural Sequence Architectures (Vaswani et al., 2017)

1. Scaled Dot-Product Attention

The fundamental building block of the Transformer is the Scaled Dot-Product Attention mechanism. Given Query (Q), Key (K), and Value (V) matrices, attention weights are computed as $\text{Attention}(Q, K, V) = \text{softmax}(Q \cdot K^T / \sqrt{d_k}) \cdot V$. The scaling factor $1/\sqrt{d_k}$ prevents dot products from growing excessively large in high dimensions, which would push softmax gradients into vanishingly small regions.

2. Multi-Head Attention

Multi-Head Attention projects Queries, Keys, and Values into h distinct representation subspaces. This allows the model to jointly attend to information from different representation subspaces at different positions. In standard architectures, 8 parallel attention heads are used, with $d_k = d_v = d_{\text{model}} / h = 64$.

3. Positional Encodings

Because Transformers contain no recurrence or convolution, order information is injected via sinusoidal positional encodings using sine and cosine functions of different frequencies. This allows the model to easily learn to attend by relative positions since $\text{PE}(\text{pos}+k)$ can be represented as a linear function of $\text{PE}(\text{pos})$.